SANDEEP SHARMA

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PROFESSIONAL SUMMARY

Lead AI Engineer with 8+ years of experience building production-grade AI systems across supply chain, manufacturing, and FMCG. Expert in end-to-end AI pipelines, LLMs, RAG, Agentic AI, and MLOps. Currently architecting an AI-powered supply chain decarbonisation platform at Mavarick, combining GPT-4o reasoning, vector search over 135 industry documents, multi-stage constraint validation, and a production FastAPI backend — delivering ranked, cited decarbonisation recommendations from raw emissions data in under 2 minutes.

CORE COMPETENCIES

- **Machine Learning & AI:** PyTorch, TensorFlow, NLP, Embeddings
- **GenAI:** LLMs, LangChain, LangGraph, RAG, Agentic AI, OpenAI, Anthropic, MCP, PEFT, LoRA
- **Programming & Tools/Libraries:** Python, R, Java, Pandas, NumPy, Scikit-learn, PySpark, AWS, SageMaker
- **LLM Models:** OpenAI, Anthropic, Google Gemini, Hugging Face, Model Context Protocol (MCP)
- **Databases:** SQL, Postgres, Redis, NoSQL, VectorDBs, FAISS, Pinecone, MongoDB Atlas, ChromaDB
- **MLOps & Automation:** Git, Docker, CI/CD, Jenkins, Kubernetes, FastAPI, uvicorn
- **Visualization & Analytics:** Power BI, Matplotlib, Seaborn, Excel
- **Other Tools:** Jira, Confluence, VSCode, PyCharm, Mathematics, Statistic

Key Skills: Analyze Large Datasets, Feature Engineering, Data mining, Predictive Modeling, Statistical Modeling, Training AI models, Anomaly detection, Data manipulation, Data pipelines, Production Deployment.

EXPERIENCE

Mavarick (March 2026 - Present)
Lead AI Engineer

- Architected and built an end-to-end AI-powered supply chain decarbonisation engine from scratch: a 4-stage pipeline (hotspot detection, lever generation, constraint validation, evaluation and ranking) that processes company emissions data and outputs ranked, actionable decarbonisation recommendations in under 2 minutes.
- Integrated GPT-4o across multiple pipeline stages: hotspot analysis with Pareto-aware AI thresholds, hybrid lever generation (deep vs. summary prompts), entity extraction for constraint validation, and a Research Agent adding market context, regulatory urgency, implementation roadmaps, and business case framing per lever.
- Built a RAG-based Knowledge Base over 135 real industry PDFs (23,170 chunks) covering EU regulations (CBAM, REACH, ELV), supplier EPDs, IPCC/IEA agency reports, and OEM sustainability disclosures. Used sentence-transformers (all-MiniLM-L6-v2) for local embedding and MongoDB Atlas vector search for citation retrieval, grounding every recommendation in real published sources.
- Designed a 133-rule constraint validation engine using GPT-4o entity extraction and 4 pure-Python validator classes (Numerical, EntityMatch, Temporal, ComplexLogic) checking every decarbonisation action against EU regulatory law, manufacturer contracts, and safety standards before surfacing it as a recommendation.
- Built and deployed a production FastAPI backend with MongoDB Atlas as the data layer for live company emissions data and saved recommendations. Implemented parallel hotspot processing via ThreadPoolExecutor, response caching, and exponential backoff retry logic for OpenAI rate limits.
- Implemented supplier anonymisation to protect commercial data before any content reaches OpenAI APIs. Built a recommendation classification engine (Quick Win, Strategic, Long-term) based on implementation timeline, CapEx, and regulatory feasibility score.

PepsiCo PGCS (July 2024 – Present)

SC DATA SCIENTIST

- Leading Data Science Entity in PepsiCo for Supply Chain Demand Planning team
- Working on open-source Large Language Models (LLMs), RAG pipelines using Langchain, Huggingface and Vector databases and used techniques like PEFT, LoRA for fine-tuning of models to create a chatbot that aligns with business needs and improve decision-making processes.
- Developed a demand forecasting tool for supply chain optimization using the combination of time-series forecasting with statistical models, machine learning, and deep learning (e.g., ARIMA, SARIMA, ETS, Propet, RF, XGBoost, LSTM, GRU).
- Selecting the best-performing model using combination of 5 metrics at the product and time granularity to get the best forecast for 30+ markets and customers.
- Increased the forecast accuracy to 82% from 55% for 15+ markets.
- Leveraged use of Copilot to create charts and semantic models in PowerBi
- Written an algorithm to automate the distribution for the optimized load distribution across the hubs with multiple thresholds and conditions

Johnson Controls (May 2022 – Feb 2024)

Data Scientist and Analyst

- Led data science and AI for fire detection SAAS products in JCI and worked closely with operations managers and other cross-functional teams, focusing on approaches like analytics, data science, and NLP with the utilization of current ML/AI techniques.
- Leveraged open-source LLM models like Mistral 7B, Falcon 40B, LangChain, VectorDB, Vector Embeddings, and RAG pipeline to create a responsive and conversational chatbot for custom JCI data. Used techniques like PEFT, LoRA for fine-tuning of models. The primary objective was to streamline and enhance the organization's operations by developing this chatbot and integrating it with the existing platform. Further, it is integrated with inventory management and real-time log analysis and decides the price of fire safety devices for commercial use.
- Responsible for creating end-to-end AI/ML projects by creating CI/CD pipelines and using techniques like MLOps and cloud system AWS.
- Automated the process of summarizing lengthy documents and reports using NLP model BART. This achieved a 60% reduction in summarization time while maintaining 95% accuracy in key information extraction.
- Created and fine-tuned the SLM BERT ML model to understand the failure reasons for installed devices using failure Notes provided by the inspectors.
- Helping in centralizing data, analyzing large datasets, creating pipelines, which can be used for delivering insights and creating visualizations to understand past trends.

Dell Technologies- July 2020 - May 2022

Senior Data Analyst & Scientist

- Worked as part of the Dell Social Media PM team, responsible for analyzing data related to performance and productivity, managing volume over different social media globally.
- Root Cause Analysis: Used RNN with pre-trained model BERT to understand the customer experience from conversations between the user and agent and identify the root cause behind the dissatisfaction of the user.
- Identifying frequent complaints: Used Topic Modeling (NLP) with LDA and LDA Mallet to determine the volume of the same type of FAQ from the complaints/questions asked by the users.
- DCF best solution: Used NLP classification techniques, bidirectional LSTM with Tf-Idf to identify whether the comment posted by the user on the Dell Community Forum is a question or simple feedback and marks the correct answer among all posts.
- Global planning and monitoring of SL, AHT, and staffing patterns for frontline agents by queue and analyzing performance with respect to goals.
- RCA of S.L. failure and opportunities so that required action can be taken to improve our performance.

- Forecasting incoming volume patterns, Experience in product pricing, Annual Operation plan by queues/ platforms/ languages. Recommend changes in volume distribution of volumes over different sites (APAC, Europe, NA, LATAM) to maximize efficiency.
- Developed an Early Warning System for highlighting Volume/ Capacity gap, to propose hiring and routing changes before we miss SL or other performance-related issues.

Tata Consultancy Services - Dec 2016 - Sep 2019
Data Scientist and Analyst

- Real-Time FAQ Chatbot: Automated the process of answering the same questions asked by customers by developing an FAQ chatbot. Used bag-of-words, Embedding, and simple as well as bi-directional LSTM techniques for Sentiment Analysis.
- Analyzed and automated reports: Developed an anomaly detection model to detect fraud for our banking domains and automated the entire process. Used a 1-class SVM ML model for outlier detection. Developed baseline decision tree, random forest, gradient boosting, and SVM models with tuned parameters.
- Developed an Artificial Intelligence-based product for the T.C.S., which was able to automate the different processes for our clients: Fault Analysis, Impact Analysis, and What-If Analysis.

EDUCATION

- **Master of Science (M.Sc.) in Data Science and Analytics MTU, Cork**
  Dates: 2019 – 2020
- **Bachelor of Technology (B.Tech) in Electrical Engineering, India**
  Dates: 2012 – 2016

CERTIFICATIONS

- Google Cloud Generative AI
- Sentiment Analysis using BERT
- SQL Certification by Hackerrank
- Data Analytics on AWS
- Visualizing Citibike Trips with Tableau
- Visual ML with Yellowbrick

KEY HIGHLIGHTS

- 2nd position in FDC Hackathon
- Runner-up in JCI Tech-Innovation

RAG Project

In johnson controls we created a rag piplines, which helped inspectors to interact with with chatbot to find the guidelines, process, instructions also to understand the root cause of the failure of the devce.
we used semantic chunking, Bert model for embadings. FAiss and chroma db as vector data base, cosine similarity and HNSW for similarity search, clobert for the reranking, and open source model like falcon 40 billion and mistral 40B as llm models

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Agentic AI Project:

For a demand planning and raw material forecasting agent within PepsiCo, a Supervisor/Hierarchical Multi-Agent Architecture is ideal, as it allows for clear specialization, sequential execution, and human-in-the-loop validation, which is critical for supply chain and financial risk.

- 🏛 Architecture and Objective
- Core Objective
- To accurately predict the future demand for finished goods and, subsequently, calculate the required raw materials and packaging inventory with optimal lead time, minimizing stockouts and excess inventory costs for PepsiCo.
- LangGraph Architecture (Supervisor Pattern)
- The entire workflow is built as a single StateGraph where a central Supervisor Agent delegates tasks to specialized sub-agents.

Component	LangGraph Role	Responsibility
State	TypedDict	A shared memory object that flows between all nodes. It stores: input_request, demand_forecast_data, material_BOM_data, raw_material_forecast, inventory_report, validation_flags, final_recommendation.
Nodes (Specialized Agents)	Python Functions/LLM Agents	Perform specific, deterministic tasks or complex reasoning steps.
Edges	Conditional Logic	Directs the workflow from one node to the next based on the current State (e.g., if a forecast confidence score is too low, route to a Human Review node).

Node Name	Agent Role	Tools/Models Used	Key Inputs (from State)	Key Outputs (to State)
1. Data Ingestion Agent	ETL Specialist	Tools: SQL_Connector (SAP, Oracle, ERP), API_Caller (external data like weather, promotions), Data_Profiler (Pandas/Polars).	input_request (e.g., product, time range)	historical_sales, metadata, external_factors
2. Demand Forecast Agent	Quantitative Analyst	Tools: Prophet_Forecaster, ARIMA_Model, LLM_Explainer.	historical_sales, external_factors	demand_forecast_data, confidence_score
3. Material Calculation Agent	BOM Specialist	Tools: BOM_Lookup (Bill of Materials data), Inventory_Check (current stock).	demand_forecast_data, BOM_data, current_inventory	raw_material_forecast, inventory_risk
4. Risk & Validation Agent	Business Analyst (LLM)	Tools: Business_Rules_Checker (Min/Max inventory), Sensitivity_Analyzer (simulations).	raw_material_forecast, inventory_risk, demand_forecast_data	validation_flags (PASS/FAIL/REVIEW), risk_summary
5. Human Review Node	Human-in-the-Loop	Tool: Send_Email/Slack_Notification (pauses the graph via LangGraph Interrupts).	All State Data	human_approval, adjusted_constraints
6. Output Agent	Reporter	Tools: PDF_Generator, ERP_Updater (write-back).	All State Data	final_recommendation